# RoleWalk: inductive structural roles for directed graphs from the characteristic function of random walks

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Draft of 01 November 2026 — results and proofs; prose to come

**Abstract.** Structural roles group nodes by their position in a graph rather than by proximity. RoleWalk describes each node by the characteristic function of its lazy random-walk transition probabilities, sampled at a few frequencies, together with the distribution of the walk's mass over distance layers and, optionally, low-order moments of a probabilistic curvature. The embedding is local, deterministic, reads directions and weights, and needs no training. We prove that it is invariant and local (hence transferable across graphs), that its distances are maximum mean discrepancies, that it is Lipschitz in edge insertions, that it counts triangles, which places it beyond 1-WL, and that distance-regular graphs with equal intersection arrays are its exact limit. Empirically it matches or beats struc2vec, GraphWave, Role2Vec, RESD and ReFeX on role classification at 10–1,000× lower cost, is best on directed roles and on transfer of centralities across graphs, and improves GPS on ZINC when combined with RWSE.

## 1 Introduction

Claims, each backed by a result below:

1. All of: inductive, directed, weighted, deterministic, online, fast (Table 1).
2. Accuracy at or above the state of the art on role classification at a fraction of the cost (Table 2, Table 11).
3. Only method that uses directions; best on synthetic and real directed roles (Table 4, Table 5).
4. Transfer: a model fitted on one graph applies to another (Table 8, Table 9; Proposition 1).
5. Expressiveness beyond 1-WL with 3 walk steps, on par with 16-step RWSE; exact characterisation of a failure class (Section 4, Table 6).
6. Useful as a structural encoding for graph transformers (Table 10).

## 2 Related work

Feature-based roles: ReFeX [1], RolX [2]. Walks and word2vec: struc2vec [3], Role2Vec [4]; directed and weighted extensions of struc2vec [5] [TODO: compare if code is available]. Diffusion: GraphWave [6], closest to RoleWalk (Proposition 10). Deep role embeddings: RESD and the RONE benchmark [7] [TODO: exact references]. Structural encodings for GNNs: RWSE [8], used by GPS [9]. Expressiveness: 1-WL [10], [11], [12], CSL [13], BREC [14]. Curvature: Ollivier-Ricci [15], [16]. Kernel mean embeddings and MMD: [17].

## 3 Method

**Walk.** $G = (V, E)$ with non-negative weights $A$ (directed or not), $\tilde{D} = \mathrm{diag}((I + A)\mathbf{1})$ and the lazy walk $H = \tilde{D}^{-1}(I + A)$. For a node $u$ and a step $w$, $p_u^w = e_u^\top H^w$ is the distribution of the walk after $w$ steps. Directed graphs use the out-walk ($A$) and the in-walk ($A^\top$) and concatenate both embeddings.

**Characteristic function.** With $f_\theta(x) = e^{i\theta x} - 1$ and a grid $\Theta = \{\theta_1, ..., \theta_m\}$ (log-spaced in $[1, 100]$, $m = 10$),

$$\varphi_u^w(\theta) = \sum_{j \in V} f_\theta\big(p_{uj}^w\big), \quad \theta \in \Theta. \tag{1}$$

Since $f_\theta(0) = 0$, the sum runs over the support of $p_u^w$ only.

**Onion layers.** With $d(u, j)$ the length of the shortest walk from $u$ to $j$, $s_u^w(k) = \sum_{j:d(u,j)=k} p_{uj}^w$ for $k = 0, ..., w$: the mass the walk keeps at each distance. Supports of earlier powers give the layers, so no extra walk is computed.

**Curvature (option).** For an edge $xy$ of the symmetrised unweighted graph, $\kappa(x,y) = \sum_z \min\big(m_x(z), m_y(z)\big)$ with $m_x$ the one-step lazy walk from $x$; $\kappa_x$ is its mean over the edges at $x$. Each step adds the mean $\mu_u^w = \sum_j p_{uj}^w \kappa_j$ and the standard deviation $\sigma_u^w$ of $\kappa$ under $p_u^w$. Since a unit of mass that does not overlap moves at least one step, $W_1\big(m_x, m_y\big) \geq 1 - \kappa(x,y)$: $\kappa$ is an upper bound of Ollivier-Ricci curvature $1 - W_1$, made of probabilities.

**Embedding.** $X_u$ concatenates, for $w = 1, ..., L$ $(L = 3)$, $\operatorname{Re}\varphi_u^w$, $\operatorname{Im}\varphi_u^w$, $\|\Theta\| s_u^w$ and, with curvature, $\|\Theta\|(\mu_u^w, \sigma_u^w)$: $2mL + L(L+3)/2(+2L) = 69$ (75) features. The factor $\|\Theta\| = \Big(\sum_\theta \theta^2\Big)^{1/2}$ puts a unit of mass in the units of $\varphi$, since $f_\theta(p) \approx i\theta p$ for small $p$; without it the layers carry 0.02% of the variance and do not change the roles found by $k$-means (Section 6.3).

**Pruning.** After each product, entries below $\text{tol} = 10^{-4}$ are dropped; their mass is added back to $\operatorname{Im}\varphi$ to first order ($\theta \cdot$ mass). Results in Section 4 are stated for $\text{tol} = 0$.

**Roles.** Ward clustering with silhouette for the number of roles, or $k$-means with a given number; fitted centroids label nodes of any graph.

# 4 Properties

Throughout, $L$ is the number of steps, $B_r(u)$ the subgraph induced by nodes within distance $r$ of $u$ (out- and in-distance for directed graphs), and statements hold for $\text{tol} = 0$.

**Proposition 1** (locality and invariance)**.** *$X_u$ is determined by the isomorphism class of the rooted graph $(B_L(u), u)$, and of $(B_{L+2}(u), u)$ with curvature. In particular it does not depend on $|V|$, nodes related by an automorphism get equal embeddings, and a node can be embedded from its neighbourhood alone.*

*Proof.* $p_{uj}^w = \sum_{u=x_0,\ldots,x_w=j} \prod_t H_{x_t x_{t+1}}$ runs over walks of length $w \leq L$ from $u$. Their nodes lie within distance $w$, and $H_{xy} = (I + A)_{xy}/\tilde{D}_{xx}$ needs the edges at $x$, with $x$ within distance $w - 1$: all are in $B_L(u)$. An isomorphism of rooted balls maps these walks bijectively and preserves their weights, so it maps $p_u^w$ to $p_{u'}^{\prime w}$ entry by entry; Equation 1 is a sum over $j$, hence equal. Distances up to $L$ are read inside $B_L(u)$. With curvature, $\kappa_j$ for $j$ within distance $L$ needs the closed neighbourhoods of $j$ and of its neighbours, inside $B_{L+2}(u)$. □

Coordinates therefore mean the same thing in every graph, which is what transfer (Table 8, Table 9) requires.

**Proposition 2** (injectivity)**.** *Let $P, Q$ be finite multisets of reals in $(0, 1]$. If $\sum_{p\in P} f_\theta(p) = \sum_{q\in Q} f_\theta(q)$ for all $\theta$ in a set with an accumulation point (e.g. $[1, 100]$), then $P = Q$. For a finite grid, given any countable family of pairs $P \neq Q$, Lebesgue-almost every $\Theta \in \mathbb{R}^m$ separates all of them.*

*Proof.* $F(\theta) = \sum_P e^{i\theta p} - \sum_Q e^{i\theta q} - (|P| - |Q|)$ is entire and vanishes on a set with an accumulation point, hence everywhere. Exponentials $e^{i\theta\lambda}$ with distinct $\lambda$ are linearly independent, so each $\lambda > 0$ has the same multiplicity in $P$ and $Q$ (and $|P| = |Q|$, coefficient of $\lambda = 0$). For the second claim, the real zeros of each nonzero entire function $F$ are countable; a countable union of countable sets is null, so almost every $\theta_1$ already separates every pair. □

So with $\Theta$ an interval, $\varphi_u^w$ is exactly the multiset of nonzero transition probabilities of $u$ after $w$ steps. The second claim is qualitative: the separation margin, not its existence, motivates $m = 10$ frequencies across two decades.

**Proposition 3** (distances are MMDs)**.** $\|\varphi_u^w - \varphi_v^w\|^2 = \text{MMD}_k^2(\mu_u, \mu_v)$ *between the counting measures $\mu_u = \sum_j \delta_{p_{uj}^w}$, for the positive-definite kernel $k(a,b) = \sum_{\theta\in\Theta}[\cos\theta(a-b) - \cos\theta a - \cos\theta b + 1]$.*

*Proof.* $\varphi_u^w = \sum_j \psi\big(p_{uj}^w\big)$ with the feature map $\psi(x) = (f_\theta(x))_{\theta\in\Theta} \in \mathbb{C}^m$, so $\varphi_u^w$ is the kernel mean embedding of $\mu_u$ and $\|\varphi_u - \varphi_v\|^2$ is the MMD for $k(a,b) = \operatorname{Re}\langle\psi(a), \psi(b)\rangle$, positive definite as an inner product. Expanding $\operatorname{Re}\big(e^{i\theta a} - 1\big)\overline{(e^{i\theta b} - 1)}$ gives the formula. Atoms at 0 have $\psi(0) = 0$: unreached nodes do not count. □

**Proposition 4** (stability)**.** *Let $G'$ be $G$ plus an unweighted edge $ab$, with $d = \min(d_a, d_b)$ the smaller endpoint degree in $G$. For every node $u$ and step $w$,*

$$\|\varphi_u^w(G') - \varphi_u^w(G)\|_2 \leq w\|\Theta\|_2 \cdot \frac{2}{d+2}, \tag{2}$$

*and the left side is 0 if $u$ is at distance $\geq w$ from both $a$ and $b$.*

*Proof.* $|e^{i\theta x} - e^{i\theta y}| \le \theta\,|x-y|$, so $\|\varphi_u^w(G') - \varphi_u^w(G)\|_2 \le \|\Theta\|_2 \|p_u'^w - p_u^w\|_1$. Telescoping, $H'^w - H^w = \sum_{t=0}^{w-1} H'^t(H'-H)H^{w-1-t}$; stochastic matrices have $\|\cdot\|_\infty = 1$, so each row of the difference has $\ell_1$ norm at most $w\|H'-H\|_\infty$. Only rows $a$ and $b$ of $H$ change; row $a$ goes from $d_a+1$ entries $1/(d_a+1)$ to $d_a+2$ entries $1/(d_a+2)$, an $\ell_1$ change of $2/(d_a+2)$. Row $u$ of $H'^t(H'-H)$ is nonzero only if a walk of $t \le w-1$ steps from $u$ reaches $a$ or $b$. □

Insertions at high-degree nodes barely move embeddings, and the effect is confined to the $(L-1)$-ball. On 200 random insertions (powerlaw-cluster graphs, 60 nodes) the largest observed move is 0.52 of the bound (`theory_checks.py`).

**Proposition 5** (degree and triangles)**.** *$s_u^1(0) = 1/(d_u+1)$ in an unweighted graph. If $G$ is $d$-regular, $s_u^3(0) = (H^3)_{uu} = (1+3d+2t_u)/(d+1)^3$, with $t_u$ the number of triangles at $u$. Hence RoleWalk with $L \ge 3$ separates 1-WL-equivalent graphs, e.g. two triangles and a hexagon, or any two $d$-regular graphs with different triangle counts.*

*Proof.* $(I+A)^3 = I + 3A + 3A^2 + A^3$ with $(A)_{uu} = 0$, $(A^2)_{uu} = d$, $(A^3)_{uu} = 2t_u$, and $H = (I+A)/(d+1)$. 1-WL gives every node of two $d$-regular graphs of the same size the same colour. □

**Proposition 6** (relation to RWSE)**.** *The distance-0 layer $s_u^w(0) = (H^w)_{uu}$ is the return probability of the lazy walk, i.e. RWSE [8] for $H$, up to $L$ steps. On a $d$-regular graph, $\{(H^w)_{uu}\}_{w \le L}$ and the simple-walk RWSE $\{(P^k)_{uu}\}_{k \le L}$, $P = A/d$, determine each other.*

*Proof.* $H = (I+dP)/(d+1)$, so $(H^w)_{uu} = (d+1)^{-w}\sum_{k=0}^{w}\binom{w}{k} d^k (P^k)_{uu}$: a triangular system with nonzero diagonal $d^w/(d+1)^w$. □

On irregular graphs the two walks differ and neither encoding contains the other; RoleWalk adds the other layers and the full distribution of transition probabilities.

**Proposition 7** (distance-regular graphs are the limit)**.** *Let $G, G'$ be distance-regular graphs with the same intersection array. Then all nodes of $G$ and $G'$ get the same embedding, for every $L$, with or without curvature. This covers strongly regular graphs with equal parameters (diameter 2).*

*Proof.* Let $A_i$ be the distance-$i$ matrices. Distance-regularity gives $AA_i = b_{i-1}A_{i-1} + a_iA_i + c_{i+1}A_{i+1}$ with coefficients from the intersection array [18], so by induction $(I+A)^w = \sum_i \alpha_{w,i}A_i$ with $\alpha_{w,i}$ depending only on the array. As $H = (I+A)/(k+1)$, $p_{uj}^w = \alpha_{w,d(u,j)}/(k+1)^w$: each row has value $\alpha_{w,i}/(k+1)^w$ with multiplicity $k_i$, the number of nodes at distance $i$, fixed by the array. So $\varphi_u^w$ and $s_u^w(i) = k_i\alpha_{w,i}/(k+1)^w$ are the same for all nodes of both graphs. For an edge $xy$, $\kappa(x,y) = |N[x] \cap N[y]|\ /(k+1) = (a_1+2)/(k+1)$, a constant, so $\mu_u^w$ is constant and $\sigma_u^w = 0$. □

Checked on Shrikhande / $4 \times 4$ rook, Chang / $T(8)$ and the diameter-3 pair Doob $D(1,1)$ / Hamming $H(3,4)$: largest difference between any two nodes $\le 3 \cdot 10^{-7}$ (float32). This explains the 0/20 distance-regular pairs of BREC (Table 7); RWSE, which is constant on these graphs for the same reason, fails them too.

**Proposition 8** (locality limits graph-level separation)**.** *Let $A$ and $B$ be a 40-cycle with pendant nodes attached at cycle positions $\{0, 15\}$ and $\{0, 20\}$. They are not isomorphic and 1-WL separates them, but RoleWalk with $L = 3$ (with curvature) does not.*

*Proof.* The pendants are more than $2(L+2)$ apart in both graphs, so every rooted ball $B_{L+2}(u)$ contains at most one pendant and is fixed by $u$'s distance to it; both graphs have the same multiset of balls. Conclude by Proposition 1. Colour refinement propagates along the cycle and eventually sees the distance between pendants (checked with 40 rounds). □

**Empirical comparison with 1-WL at equal depth.** A walk of $L$ steps sees degrees up to distance $L-1$, as does 1-WL after $L$ rounds from uniform colours. Over all 6,781 nodes of the 996 connected graphs with at most 7 nodes, the partition induced by RoleWalk ($L = 3$) refines that of 3-round 1-WL: no two nodes that 1-WL separates share an embedding, and RoleWalk splits 677 of the 1-WL classes (`theory_checks.py`).

**Conjecture 9.** *For every $L$, the partition of nodes (across all graphs) induced by RoleWalk with $L$ steps and $\Theta$ an interval refines the partition of $L$-round 1-WL.*

Together with Proposition 5 and the propositions above: at equal depth RoleWalk is strictly finer than 1-WL on all graphs up to 7 nodes, while being local, which 1-WL with unbounded rounds is not.

**Proposition 10** (relation to GraphWave)**.** *GraphWave [6] embeds $u$ by $\chi_u(t) = \frac{1}{n}\sum_j e^{it\Psi_{ju}}$ with $\Psi = Ug_s(\Lambda)U^\top$ the heat kernel at a scale $s$ chosen from the spectrum of each graph. Then $n(\chi_u(t) - 1) = \sum_j f_t(\Psi_{ju})$: Equation 1 with the heat kernel in place of $H^w$. $\chi_u$ depends on $n$ and on $s$, so the same rooted neighbourhood gets different coordinates in graphs of different sizes or spectra.*

*Proof.* $\sum_j \left(e^{it\Psi_{ju}} - 1\right) = n\chi_u(t) - n$. $\Psi$ has full support on a connected graph, so Proposition 1 fails; $1/n$ and $s$ are graph-level. □

This is why GraphWave transfers at chance level (Table 9), while RoleWalk keeps its accuracy.

**Proposition 11** (cost)**.** *With pruning, each row of each power keeps at most $1/\operatorname{tol}$ entries, so step $w$ costs $O\left(\sum_u \sum_{j \in \operatorname{supp} p_u^{w-1}} (d_j + 1)\right)$ with $|\operatorname{supp} p_u^{w-1}| \le \min(|B_{w-1}(u)|, 1/\operatorname{tol})$; the characteristic function adds $O(m \cdot \operatorname{nnz})$ and the layers $L$ element-wise products. Curvature costs $O\left(\sum_u d_u^2\right)$. Embedding a set $S$ touches only $B_L(S)$ ($B_{L+2}(S)$ with curvature).*

*Proof.* Entries are non-negative and each row sums to at most 1, so at most $1/\operatorname{tol}$ of them are $\ge \operatorname{tol}$. The rest follows from sparse products and Proposition 1. □

# 5 Experiments

All CPU timings on one core. Classification: logistic regression on standardised embeddings, 50 stratified 80/20 splits. Best in bold, second underlined. Code and CSV files in `benchmarks/`.

Table 1: Properties. Checked against each paper; "~" partial.

| | Transfer | Directed | Weighted | Time (USA) | Online | Deterministic |
|---|---|---|---|---|---|---|
| **RoleWalk** | yes | yes | yes | 0.2 s | yes | yes |
| struc2vec | no | no[a] | no[a] | 564 s | no | no |
| GraphWave | no (P. 10) | ~ | yes | 18 s | no | yes |
| Role2Vec | ~ | yes[b] | no | 10 s | ~ | no |
| RESD | no | no | no | 8 s | no | no |
| ReFeX | ~ | ~ | ~ | 0.1 s | ~ | yes |
| RWSE | yes | ~ | yes | 1.7 s | yes | yes |

[a] struc2vec++ [5] extends it. [b] In the paper; the reference implementation is undirected. [TODO: verify RESD, ReFeX, RWSE rows]

Table 2: Role classification accuracy. struc2vec is unseeded: mean ± std over 5 runs. `roles.csv`, `struc2vec_runs.csv`.

| | Brazil | Europe | USA | Reality-call | Actor | Film* |
|---|---|---|---|---|---|---|
| **RoleWalk** | 0.743 | <u>0.583</u> | <u>0.657</u> | **0.860** | **0.487** | 0.569 |
| **RoleWalk + curv.** | 0.741 | **0.596** | **0.666** | **0.860** | <u>0.485</u> | <u>0.581</u> |
| struc2vec | 0.705 ± .022 | 0.561 ± .020 | 0.596 ± .004 | – | – | – |
| RESD | 0.728 | 0.544 | 0.653 | 0.857 | 0.476 | 0.563 |
| GraphWave | **0.769** | 0.561 | 0.587 | **0.860** | 0.481 | – |
| ReFeX | 0.691 | 0.576 | 0.622 | 0.851 | 0.478 | 0.579 |
| Role2Vec | 0.476 | 0.366 | 0.562 | 0.844 | 0.399 | **0.743** |
| RWSE | 0.711 | 0.572 | 0.517 | 0.855 | 0.363 | 0.450 |
| Degree | <u>0.756</u> | 0.554 | 0.567 | <u>0.858</u> | 0.476 | 0.478 |

* A node-type task (Section 6.4). "–": too slow or out of memory.

Table 3: Wikipedia talk networks (RONE): precision@10 for bots and admins, mean over networks. `roles.csv`.

| | Bots | Admins |
|---|---|---|
| RoleWalk | 0.129 | 0.324 |
| struc2vec (3 smallest networks) | **0.146** | **0.391** |
| GraphWave | 0.115 | 0.298 |
| RESD | 0.114 | 0.254 |
| ReFeX | 0.101 | 0.299 |
| RWSE | 0.118 | 0.297 |
| Degree | 0.075 | 0.296 |

Mixed. Networks are distributed undirected. [TODO: directed versions from SNAP; per-network table]

Table 4: Directed roles on synthetic graphs (directed motifs hung from a directed cycle; several roles coincide once directions are dropped). 3 seeds; noise = share of edges rewired. `directed.csv`.

| | Acc. 0% | Acc. 5% | Acc. 10% | ARI 5% | mAP 5% |
|---|---|---|---|---|---|
| **RoleWalk + curv.** | **0.989** | **0.712** | **0.547** | **0.381** | **0.625** |
| RoleWalk | 0.908 | 0.680 | 0.510 | 0.366 | 0.574 |
| RoleWalk, undirected | 0.767 | 0.560 | 0.420 | 0.351 | 0.479 |
| Role2Vec | 0.730 | 0.451 | 0.368 | 0.148 | 0.295 |
| ReFeX | 0.532 | 0.429 | 0.431 | 0.342 | 0.356 |
| GraphWave | 0.586 | 0.388 | 0.383 | 0.317 | 0.416 |
| RESD | 0.547 | 0.420 | 0.423 | 0.253 | 0.348 |
| struc2vec | 0.510 | 0.366 | 0.277 | 0.141 | 0.260 |
| In/out degree | 0.516 | 0.466 | 0.405 | 0.310 | 0.349 |

Table 5: Real directed roles: Wikipedia requests for adminship [19] (11,377 users, 188k votes voter → candidate). A: elected among the 3,494 candidates. B: voter / failed candidate / admin. `roles_rfa.csv`.

| | A: elected | B: 3 classes | Time |
|---|---|---|---|
| **RoleWalk + curv.** | **0.823** | **0.943** | 5 s |
| RoleWalk | 0.808 | 0.938 | 4 s |
| RoleWalk, undirected | 0.791 | 0.863 | |
| RoleWalk + curv., undirected | 0.814 | 0.900 | |
| RESD | 0.798 | 0.843 | 68 s |
| Role2Vec | 0.798 | 0.857 | 82 s |
| ReFeX | 0.794 | 0.847 | 1 s |
| In/out degree | 0.755 | 0.922 | 0 s |
| RWSE (out-walk) | 0.690 | 0.826 | 51 s |

struc2vec and GraphWave do not run at this size.

Table 6: Expressiveness: pairs told apart. Every set is 1-WL-indistinguishable. No false difference on random relabellings. `expressiveness.csv`.

| | Atlas ≤ 7 (20) | Regular (3,181) | CSL (10 cl.) | Decalin / bicyc. | $2C_3$ / $C_6$ | SRG pair* |
|---|---|---|---|---|---|---|
| **RoleWalk (3 steps)** | **20** | **3,181** | 6 | yes | yes | no |
| RoleWalk (5 steps) | **20** | **3,181** | **10** | yes | yes | no |
| RoleWalk, no layers | 16 | 3,180 | 6 | yes | yes | no |
| RWSE (16 steps) | **20** | 3,180 | **10** | yes | yes | no |
| Degree (= 1-WL) | 0 | 0 | 1 | no | no | no |

* Shrikhande / $4 \times 4$ rook, beyond 3-WL; not separable by Proposition 7.

Table 7: BREC [14] (400 pairs).

| | Basic (60) | Regular (100) | Extension (100) | CFI (100) | 4-vertex (20) | Dist.-reg. (20) | Total |
|---|---|---|---|---|---|---|---|
| RoleWalk (3 steps) | 60 | 50 | 97 | 0 | 0 | 0 | 207 |
| RoleWalk (5 steps) | 60 | 50 | 100 | 3 | 0 | 0 | 213 |
| RoleWalk, no layers | 58 | 48 | 97 | 0 | 0 | 0 | 203 |
| RWSE (16 steps) | 60 | 50 | 100 | 4 | 0 | 0 | **214** |

Distance-regular: 0 by Proposition 7. The 50 regular pairs missed are exactly the 50 strongly regular pairs, all with equal parameters (checked).

Table 8: Centrality transfer: a gradient-boosting regressor from embedding to centrality is fitted on one graph and evaluated (Spearman) on another graph of the same family and size (other seed) (Barabási-Albert, powerlaw-cluster, Watts-Strogatz, SBM, random geometric, motifs). `centrality_transfer.csv`.

| | RoleWalk | + curv. | ReFeX† | RWSE | GraphWave | Degree |
|---|---|---|---|---|---|---|
| Betweenness | 0.901 | **0.905** | 0.844 | 0.708 | 0.723 | 0.634 |
| Closeness | **0.866** | 0.864 | 0.684 | 0.732 | 0.597 | 0.428 |
| Clustering | **0.873** | 0.870 | 0.852 | 0.803 | 0.392 | 0.212 |
| k-core | 0.747 | 0.748 | **0.935** | 0.660 | 0.566 | 0.692 |
| Eigenvector | **0.656** | 0.649 | 0.492 | 0.531 | 0.462 | 0.428 |
| Harmonic | **0.918** | 0.916 | 0.782 | 0.784 | 0.699 | 0.500 |
| PageRank | **0.990** | **0.990** | 0.968 | 0.864 | 0.857 | 0.927 |
| Mean | **0.850** | 0.849 | 0.794 | 0.726 | 0.614 | 0.546 |

† Only where its number of features matches across graphs.

Table 9: Role transfer between airport networks: classifier fitted on one network, tested on another (6 ordered pairs, accuracy). Labels are activity quartiles, close to degree. `role_transfer.csv`.

| Degree | RoleWalk | + curv. | RoleWalk, no layers | RWSE | GraphWave |
|---|---|---|---|---|---|
| 0.542 | **0.488** | 0.470 | 0.445 | 0.375 | 0.249 (chance) |

Table 10: Structural encodings for GNNs on ZINC (test MAE at the best validation epoch). Top: GINE, 4 layers, ≈230k parameters, 300 epochs, 3 seeds. Bottom: GPS [9] protocol, 10 layers, ≈470k parameters, 500 epochs, 2 seeds. `pe_zinc_v13.csv`, `pe_zinc_gps.csv`.

| GINE | Test MAE | |
|---|---|---|
| none | 0.202 ± 0.010 | |
| RWSE | 0.107 ± 0.004 | |
| RoleWalk | 0.108 ± 0.008 | |
| RoleWalk + RWSE | 0.104 ± 0.006 | |
| **RoleWalk + curv. + RWSE** | **0.095 ± 0.002** | |
| **GPS** | **Test MAE** | **Val. MAE** |
| RWSE (published: 0.070 ± 0.004) | 0.077 ± 0.006 | 0.103 ± 0.000 |
| RoleWalk + curv. | 0.076 ± 0.001 | 0.092 ± 0.003 |
| **RoleWalk + curv. + RWSE** | **0.073 ± 0.001** | **0.089 ± 0.005** |

[TODO: GPS seeds 2-3 running; Peptides (LRGB)]

Table 11: Embedding time against size, one thread, defaults (tol $10^{-4}$, 3 steps, layers). `scalability.csv`.

| Nodes | 10 k | 30 k | 100 k | 300 k | 1000 k |
|---|---|---|---|---|---|
| Barabási-Albert ($m = 3$) | 1.11 s | 3.72 s | 15.16 s | 63.8 s | 254.56 s |
| + curvature | 1.19 s | 3.87 s | 16.61 s | 55.29 s | – |
| Watts-Strogatz ($k = 6$) | – | – | – | – | – |
| + curvature | – | – | – | – | – |

Peak memory 3.7 GB at 300k nodes (blocks of 50k rows). For comparison, at 1,190 nodes (USA airports): struc2vec 564 s, GraphWave 18 s, Role2Vec 10 s, RESD 8 s; struc2vec and GraphWave do not run on wiki-RfA (11k nodes).

# 6 Analysis

## 6.1 Summaries of the walk distribution

Table 12: Same budget of features, replacing the characteristic function. Airports: mean accuracy; Transfer: airport transfer; Motifs: synthetic accuracy.

| | Airports | Transfer | Motifs (varied / houses) | Time 20k |
|---|---|---|---|---|
| **Characteristic function** | 0.656 | 0.445 | **0.744** / 0.733 | 6.5 s |
| Laplace transform | **0.657** | 0.345 | 0.614 / 0.717 | 18 s |
| Dyadic histogram (Haar) | 0.631 | 0.353 | 0.614 / 0.725 | 15 s |
| Power sums (Rényi) | 0.653 | 0.420 | 0.647 / 0.692 | 9.7 s |
| Top-10 probabilities | 0.652 | **0.512** | 0.679 / **0.758** | 3.5 s |

## 6.2 Geometry: moments rather than integrated characteristic functions

Table 13: Ways to add local geometry. Anything that depends on graph size (degree marks, counting bounds) hurts transfer.

| | Airports | Transfer | Motifs (varied / houses) | Directed acc. |
|---|---|---|---|---|
| No geometry | 0.656 | 0.445 | 0.744 / 0.733 | 0.678 |
| + onion layers (moments) | 0.661 | 0.488 | 0.744 / 0.758 | 0.682 |
| + layers, one CF per layer | 0.654 | 0.424 | 0.751 / 0.742 | 0.695 |
| + degree marks (joint CF) | 0.65–0.67 | 0.34–0.36 | – | – |
| + curvature, counting bound | 0.667 | 0.388 | **0.754** / **0.767** | 0.721 |
| + curvature, probabilistic (moments) | **0.669** | 0.470 | 0.725 / 0.742 | 0.712 |
| + curvature, exact Ollivier-Ricci | 0.662 | **0.494** | 0.746 / 0.750 | **0.743** |
| + curvature, integrated CF | 0.663 | 0.400 | 0.751 / 0.750 | 0.697 |

The probabilistic bound is kept over exact Ollivier-Ricci for cost (no optimal transport) at a small loss.

## 6.3 Kernel scaling

In raw embeddings $\theta = 100$ carries 53% of the variance; unscaled layers carry 0.02% and leave $k$-means roles unchanged. A grid of 48 configurations (frequency weighting, range, steps, layers), scored on synthetic and real networks with equal weight (`kernel_grid.csv`): scaling the layers by $\|\Theta\|$ is what matters (score 0.274 $\rightarrow$ 0.323), the frequency weighting barely does, and the defaults win whichever dataset is left out.

## 6.4 Film is a node-type task

Table 14: Film (RONE). Class 3 is an independent set in a separate block of node ids.

| | Class 3 vs rest | Within classes 0–2 |
|---|---|---|
| Role2Vec | **0.946** | 0.637 |
| RoleWalk + curv. | 0.740 | 0.634 |
| ReFeX | 0.745 | 0.627 |
| Degree | 0.536 | 0.623 |
| Majority class | 0.593 | 0.623 |

Classes 0–2 are not predictable by any method; edge homophily is 0.28, below the 0.33 of random labels. Role2Vec's lead comes from walk co-occurrence, i.e. position, which an automorphism-invariant embedding cannot capture by Proposition 1. Film should not rank role methods.

# 7 Limitations

- Walk statistics cannot separate distance-regular graphs with equal arrays (Proposition 7), CFI pairs, or graphs whose balls coincide (locality).
- Middle-tier roles on real networks often form a continuum; per-node stability should be reported next to roles.
- Degree-like labels (airports) favour trivial baselines.

- One real labelled directed benchmark.
- GPS on ZINC: 2 seeds; the test gain is within published seed noise, clearer on validation.

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